

1. Log in to Cisco Packet Tracer or any network simulation tool, add a switch, and create two VLANs named 'Sales' and 'Support' with VLAN IDs 10 and 20.

The screenshot displays the Cisco Packet Tracer interface with a switch configuration window open. The configuration includes creating two VLANs, 'Sales' (VLAN 10) and 'Support' (VLAN 20), and assigning ports to them. The switch is named 'Switch0' and is configured with the following commands:

```
Switch0#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch0(config)#vlan 10
Switch0(config-vlan)#name sales
Switch0(config-vlan)#vlan 20
Switch0(config-vlan)#name support
Switch0(config-vlan)#exit
Switch0(config)#interface range fa0/1-3
Switch0(config-if-range)#switchport mode access
Switch0(config-if-range)#switchport access vlan 10
Switch0(config-if-range)#exit
Switch0(config)#interface range fa0/4-5
Switch0(config-if-range)#switchport mode access
Switch0(config-if-range)#switchport access vlan 20
Switch0(config-if-range)#exit
Switch0(config)#interface g0/1
Switch0(config-if)#switchport mode trunk
Switch0(config-if)#exit
Switch0(config)#exit
Switch0#
```

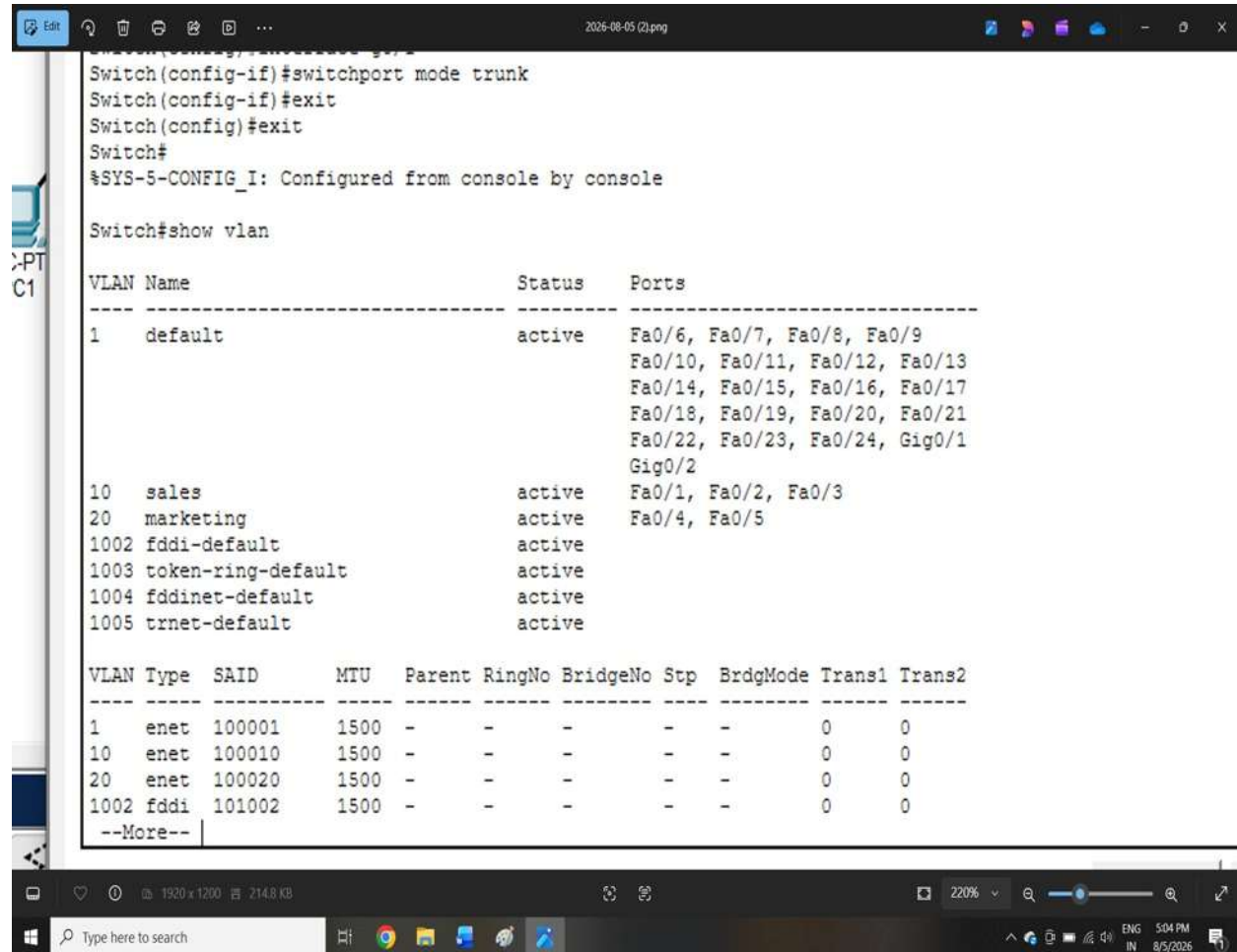
The configuration is saved to the startup configuration. The output of the 'show vlan' command is as follows:

VLAN Name	Status	Ports
1 default	active	Fa0/4, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
10 sales	active	Fa0/1, Fa0/2, Fa0/3
20 support	active	Fa0/4, Fa0/5
1002 fddi-default	active	
1005 token-ring-default	active	
1008 fddi-net-default	active	
1005 trnet-default	active	

The output of the 'show vlan detail' command is also displayed:

VLAN Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BridgeMode	Trans1	Trans2
1	enat	100001	1500	-	-	-	-	0	0
10	enat	100010	1500	-	-	-	-	0	0
20	enat	100020	1500	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	0	0

2. Assign three different switch ports to VLAN 10 (Sales) and two ports to VLAN 20 (Support), then verify the VLAN membership using the show vlan command. Hint: Use the interface range command to select multiple ports efficiently.



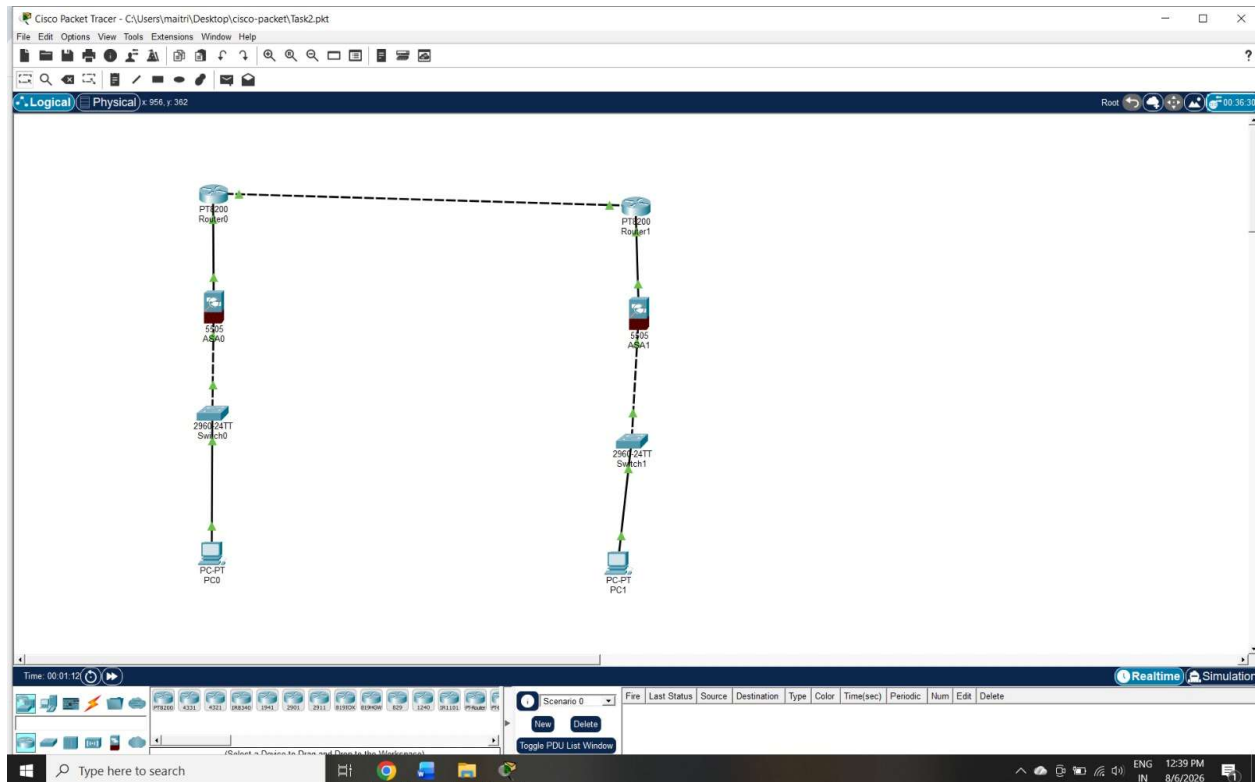
```
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan

VLAN Name                Status    Ports
-----
1    default                active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                           Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                           Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                           Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                           Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                           Gig0/2
10   sales                  active    Fa0/1, Fa0/2, Fa0/3
20   marketing              active    Fa0/4, Fa0/5
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active

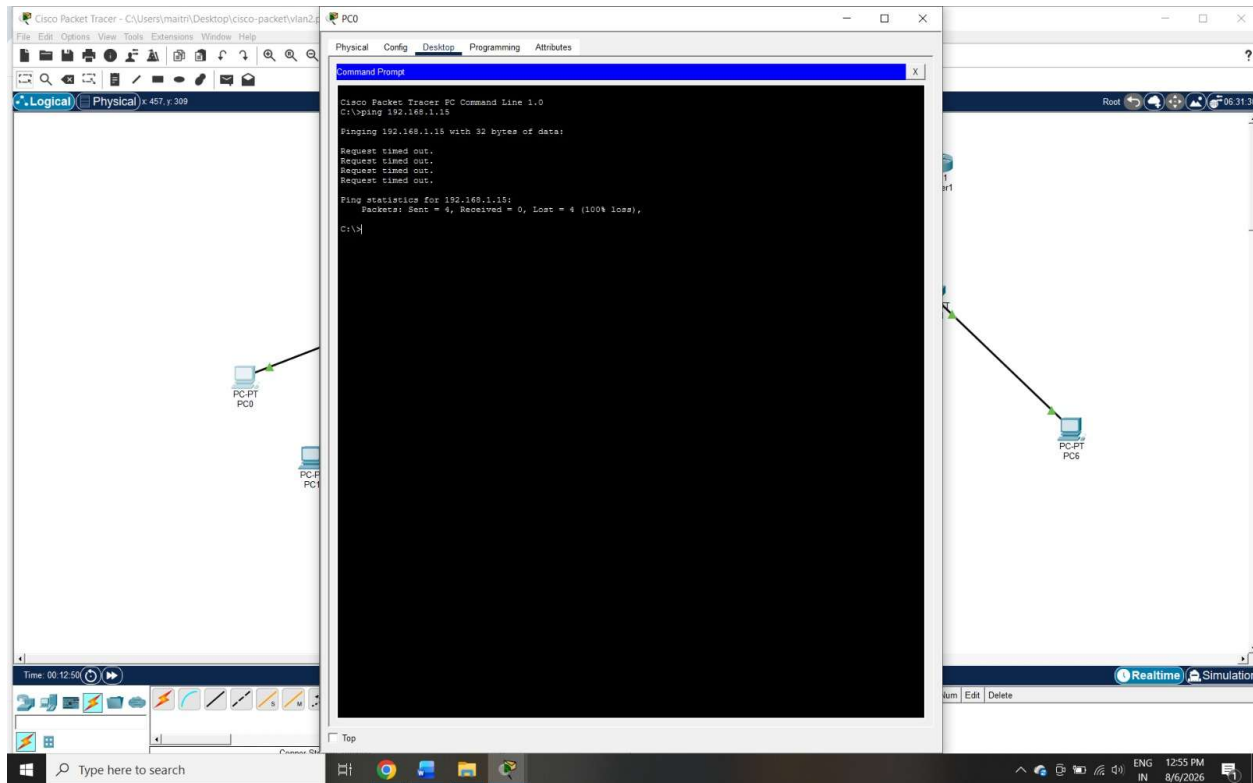
VLAN Type  SAID      MTU    Parent RingNo BridgeNo Stp    BrdgMode Trans1 Trans2
-----
1    enet     100001    1500    -      -      -      -      -      0      0
10   enet     100010    1500    -      -      -      -      -      0      0
20   enet     100020    1500    -      -      -      -      -      0      0
1002 fddi     101002    1500    -      -      -      -      -      0      0
--More--
```

3. Connect two routers using a serial or Ethernet link in your simulation tool, assign IP addresses to their interfaces, and configure a static route on Router 1 to reach the LAN behind Router 2.



4. Simulate a scenario where a device in the 'Sales' VLAN tries to ping a device in the 'Support' VLAN and explain why the ping fails, then suggest what configuration would allow inter-VLAN communication.

Hint: Think about router-on-a-stick or Layer 3 switch concepts.



The message **"Request timed out"** means that **PC0 sent a ping request, but it did not receive a reply from the destination device (192.168.1.15)** within the allowed time.